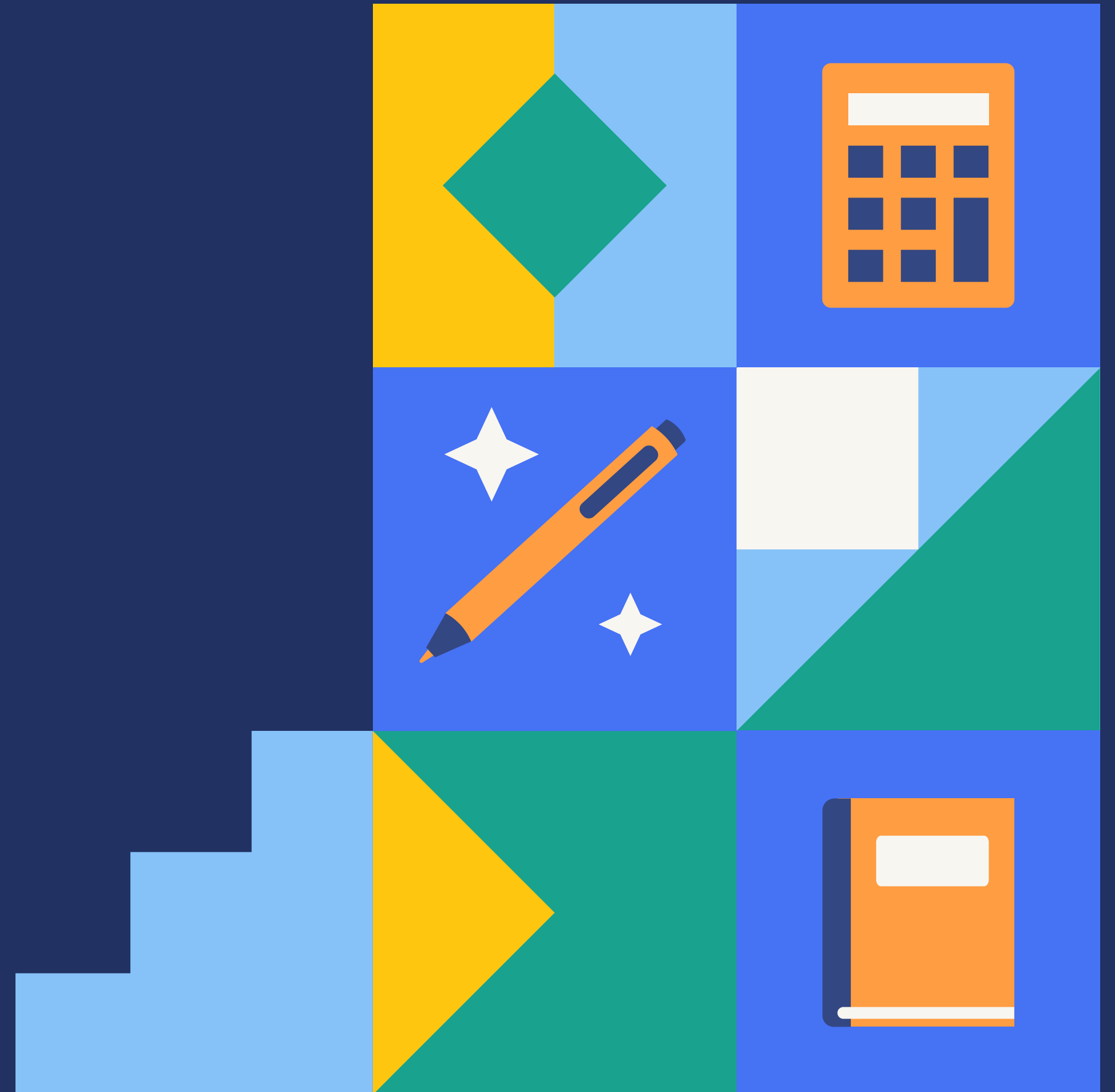
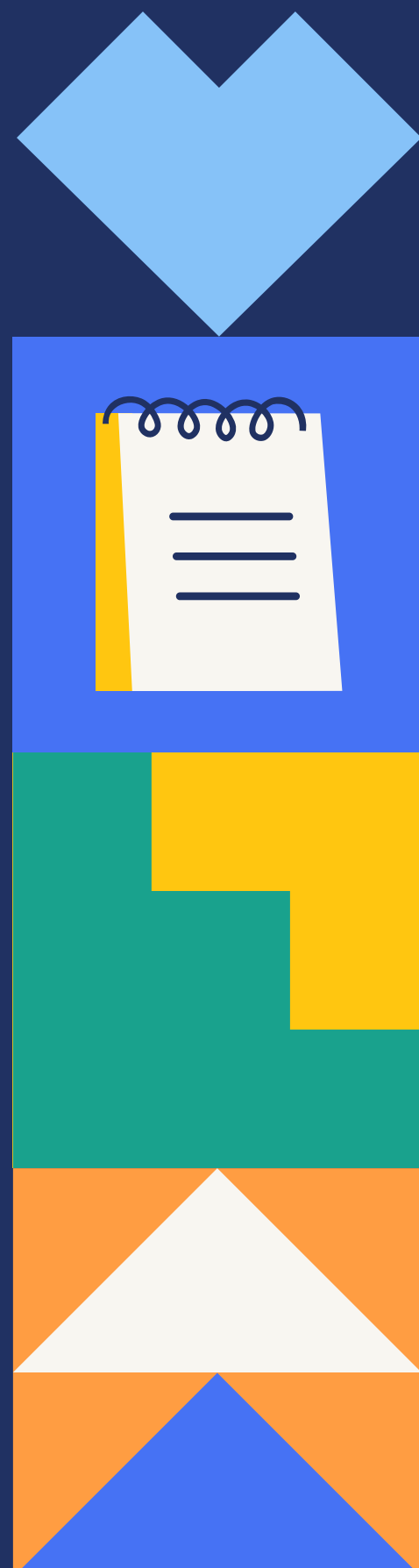


ANOVA TEST

Analysis of Variance

Presented by
Hina Ali





Lesson Outline



Presenting Results

- Frequency
- Mode

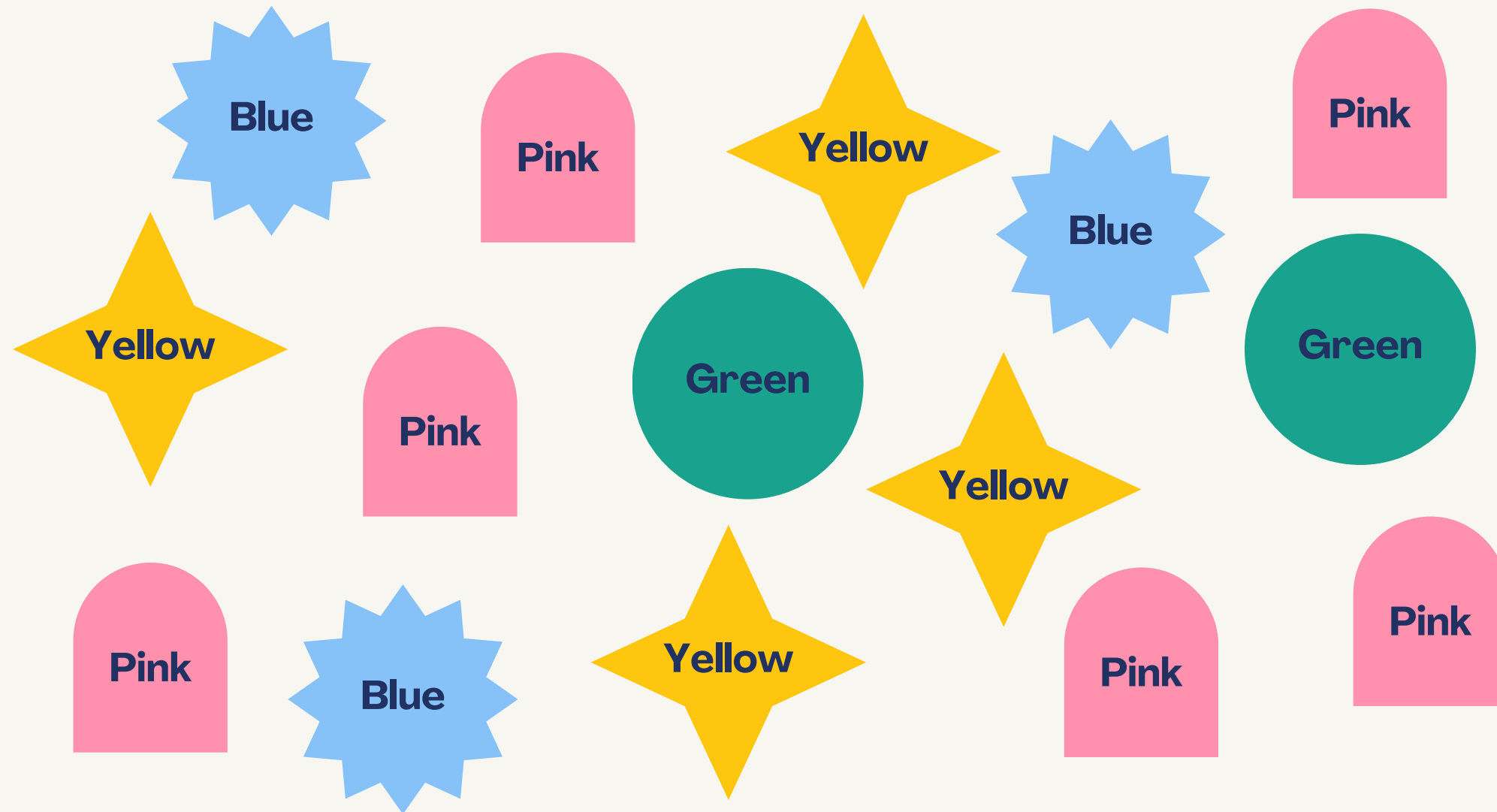


Summarizing Results

- Mean
- Median

Favorite Color Poll

Jan asks 15 of her friends what their favorite color is. Here are the answers she gathered:



How can Jan organize and present the results?



By the end of the lesson, you'll be able to:



Summarize data using frequency, mode, mean, and median.



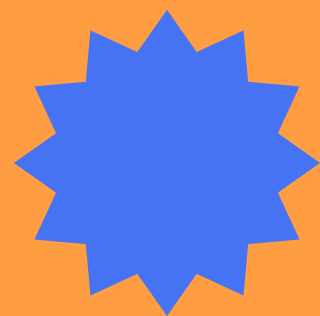
Present data using a frequency table.



Create a conclusion from the data.



Presenting Results



Frequency

refers to the number of times a result appears in the data set



Mode

refers to the result with the greatest frequency



Presenting Results



Frequency Table

It is used to represent the frequency of different results in the data.

result or
data value



frequency



Color

Number of Friends



Presenting Results

The equivalent frequency of Jan's poll is shown below:

Color	Number of Friends
Pink	6
Green	2
Blue	3
Yellow	4



Which color is the mode?



Presenting Results

The equivalent frequency of Jan's poll is shown below:

Color	Number of Friends
Pink	6
Green	2
Blue	3
Yellow	4

Pink appears most often in the poll. Therefore, it is the **mode**.

Remember: A data set may have one mode, multiple modes, or no mode.

Answer Key

Try This!

1. The modes of the data set are 2 and 5.
2. There are 12 students in the class.



modes:
2 and 5

Score	Frequency
2	3
5	3
7	1
12	1
13	2
14	2

Total: 12

Summarizing Results

Two measures of center can be used to summarize numerical data sets: the mean and the median.



Mean

average of all the
data values in the set

$$\text{Mean} = \frac{\text{sum of all data values}}{\text{number of data values}}$$



Median

the "middlemost"
value in the set



Summarizing Results

Find the mean of the following data set: 8, 3, 4, 9, 6

$$\text{Mean} = \frac{\text{sum of all data values}}{\text{number of data values}}$$

$$\text{Mean} = \frac{8 + 3 + 4 + 9 + 6}{5}$$

$$\text{Mean} = \frac{30}{5} = 6$$

The mean or average of the data set is 6.





Summarizing Results

Finding the median changes depending on the number of values in the data set.

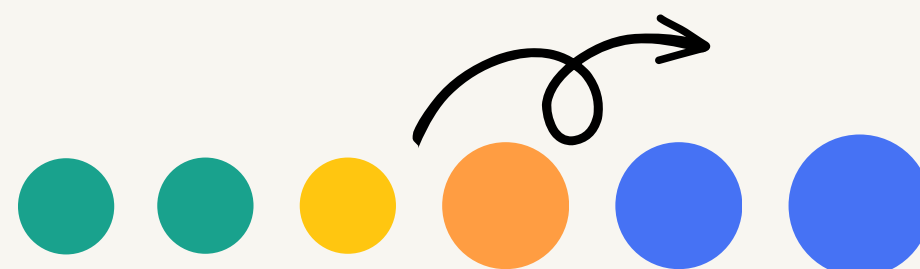
Arrange the values from least to greatest.

least $\longrightarrow$ greatest

If odd number of values:



If even number of values:



Median: $\frac{\text{yellow circle} + \text{orange circle}}{2}$



Summarizing Results

Find the median of the following data sets:

Set A: 8, 3, 4, 9, 6

3, 4, 6, 8, 9



Median

Set B: 11, 17, 3, 14, 19, 7

3, 7, 11, 14, 17, 19

$$\text{Median} = \frac{11 + 14}{2} = 12.5$$

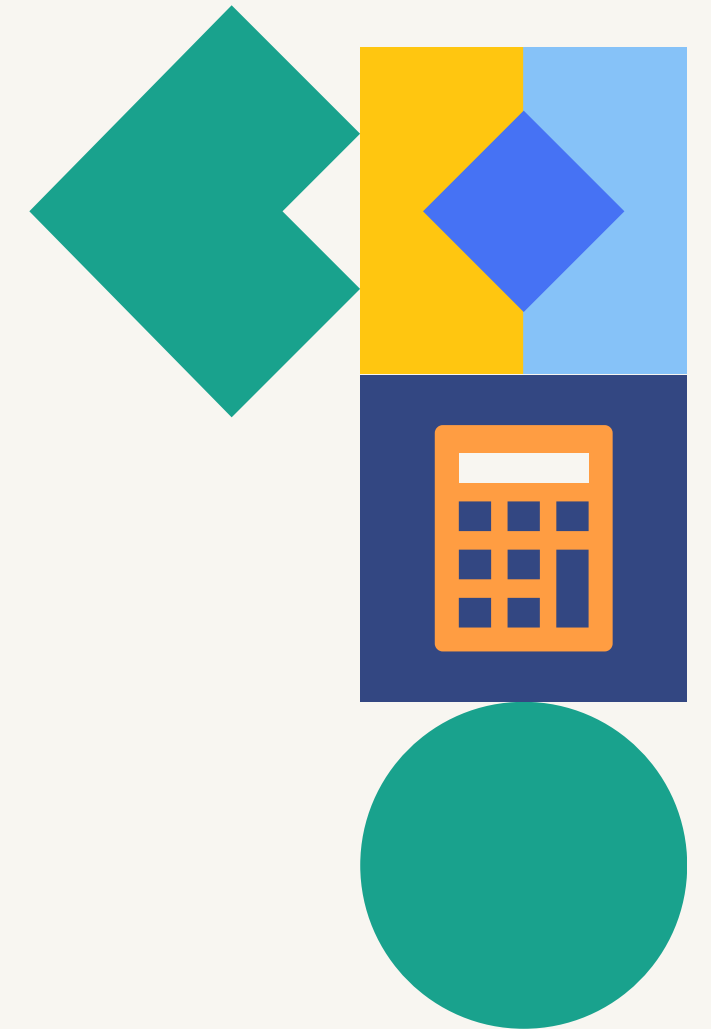


Try This!

Which two data sets have the same mean?

- A 9, 11, 15, 21, 7
- B 12, 11, 14, 15, 9
- C 7, 11, 30, 8, 9
- D 6, 14, 10, 13, 22

Find also the median for all four data sets.



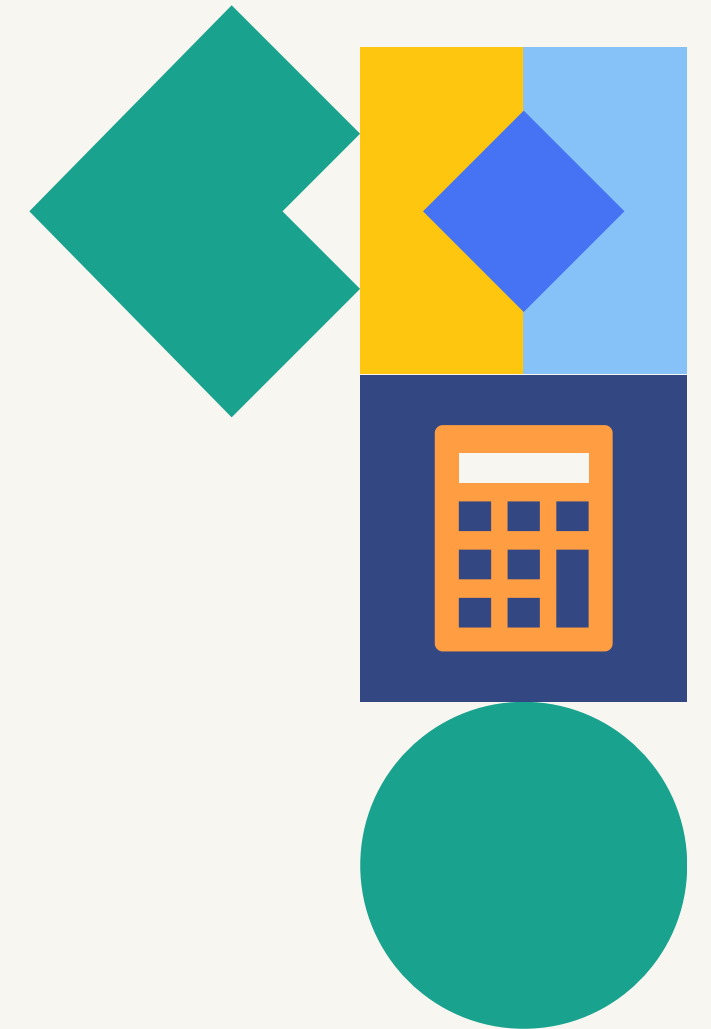
Answer Key

Try This!

Which two data sets have the same mean?

A	9, 11, 15, 21, 7	Mean: 12.6	7, 9, 11, 15, 21	Median: 11
B	12, 11, 14, 15, 9	Mean: 12.2	9, 11, 12, 14, 15	Median: 12
C	7, 11, 30, 8, 9	Mean: 13	7, 8, 9, 11, 30	Median: 9
D	6, 14, 10, 13, 22	Mean: 13	6, 10, 13, 14, 22	Median: 13

C and D have the same mean of 13.



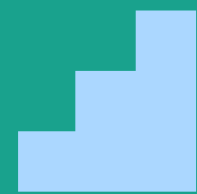


Review



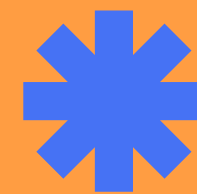
Frequency

number of times
a result appears
in the data set



Mode

result with
the greatest
frequency



Mean

average of all
the data values
in the set



Median

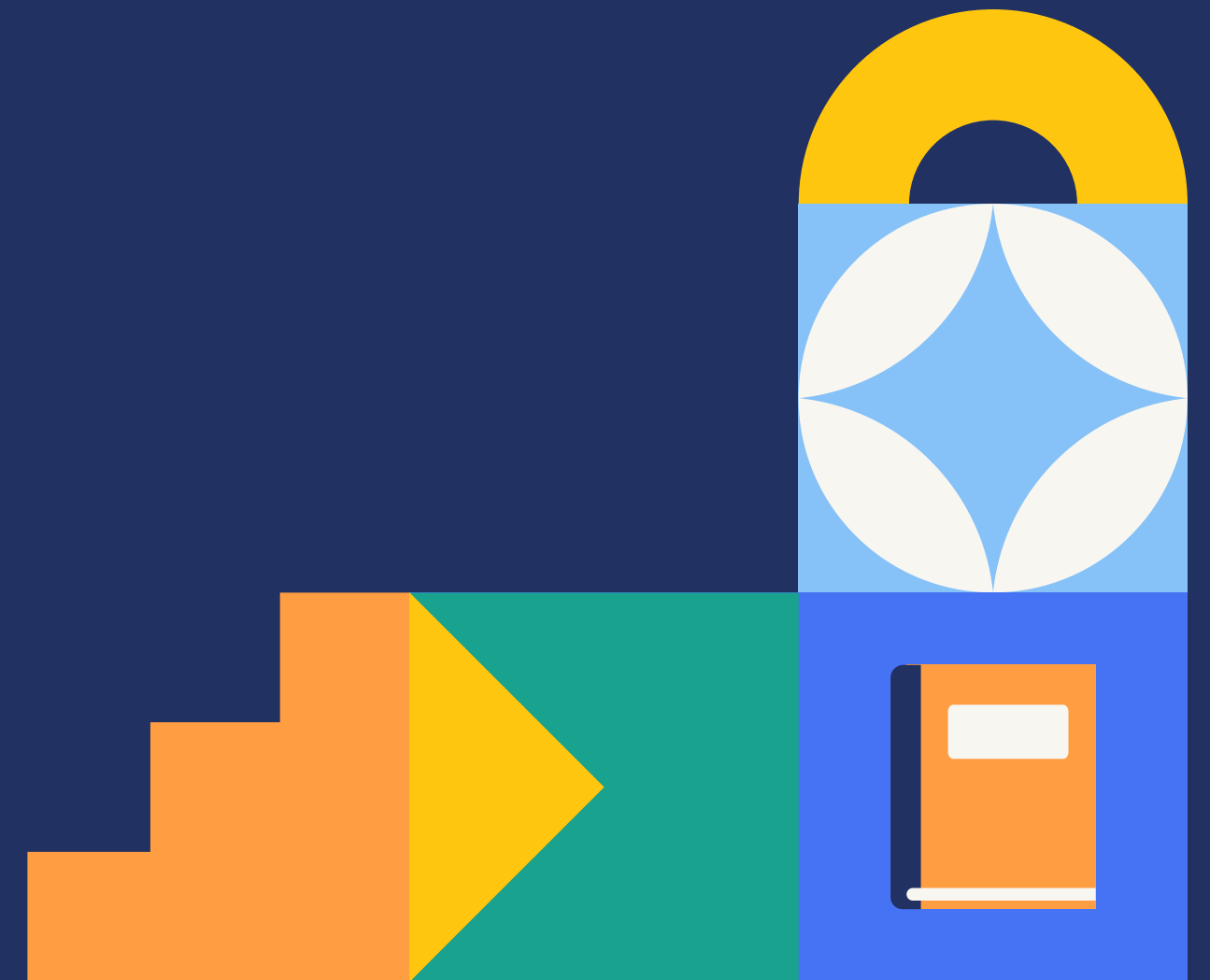
the
"middlemost"
value

Assignment

Design a survey for classmates. Choose one question from the provided options or create your own.

- What are your preferred sports?
- How many pets do you have at home?
- What is your mode of transportation when traveling to school?

Organize, summarize, and present the data you collected in a poster.





Reference

Mathspace. "Statistics." Accessed 14 June 2023,
<https://mathspace.co/textbooks/syllabuses/Syllabus-1155/topics/Topic-21893/subtopics/Subtopic-279788/?searchString=&activeTab=worksheet>.

Resource Page

Use these icons and illustrations in your Canva Presentation. Happy designing! Don't forget to delete this page before presenting.





Try this background for online class.
*Please delete this section before downloading.



**Press these
keys while
on Present
mode!**



for blur



for a drumroll



for bubbles



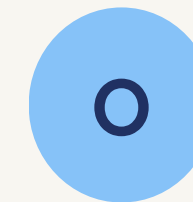
for unveil



for confetti



for mic drop



for quiet



Any number from
0-9 for a timer